

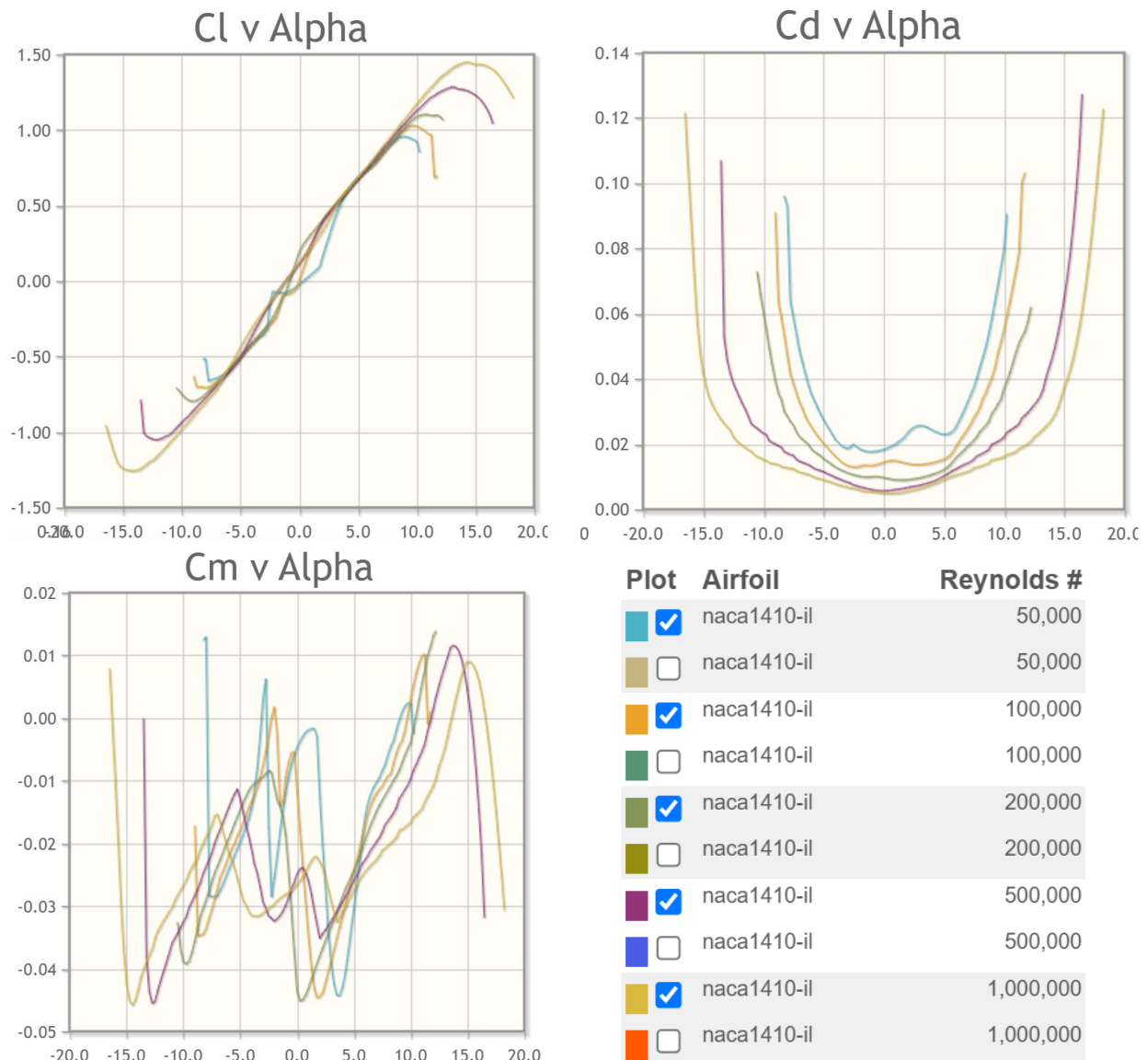
Project 1: Airfoil Analysis

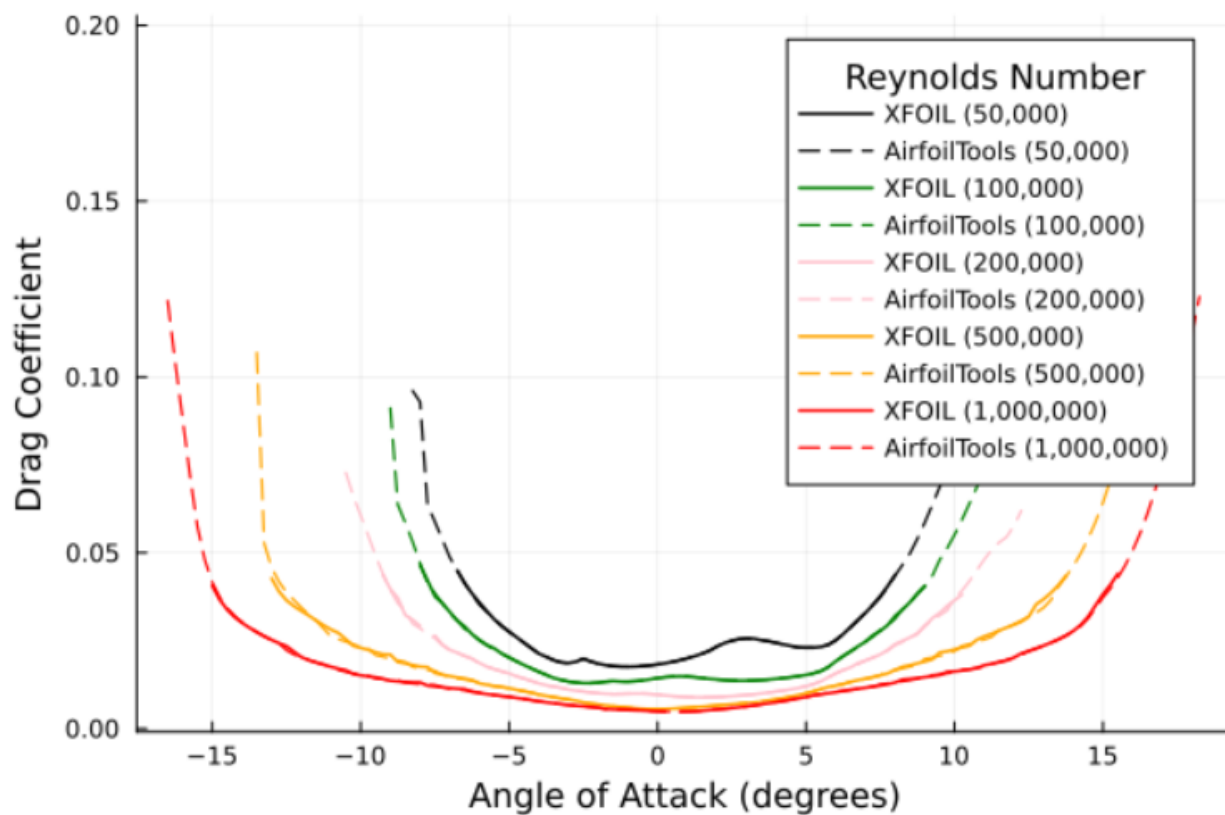
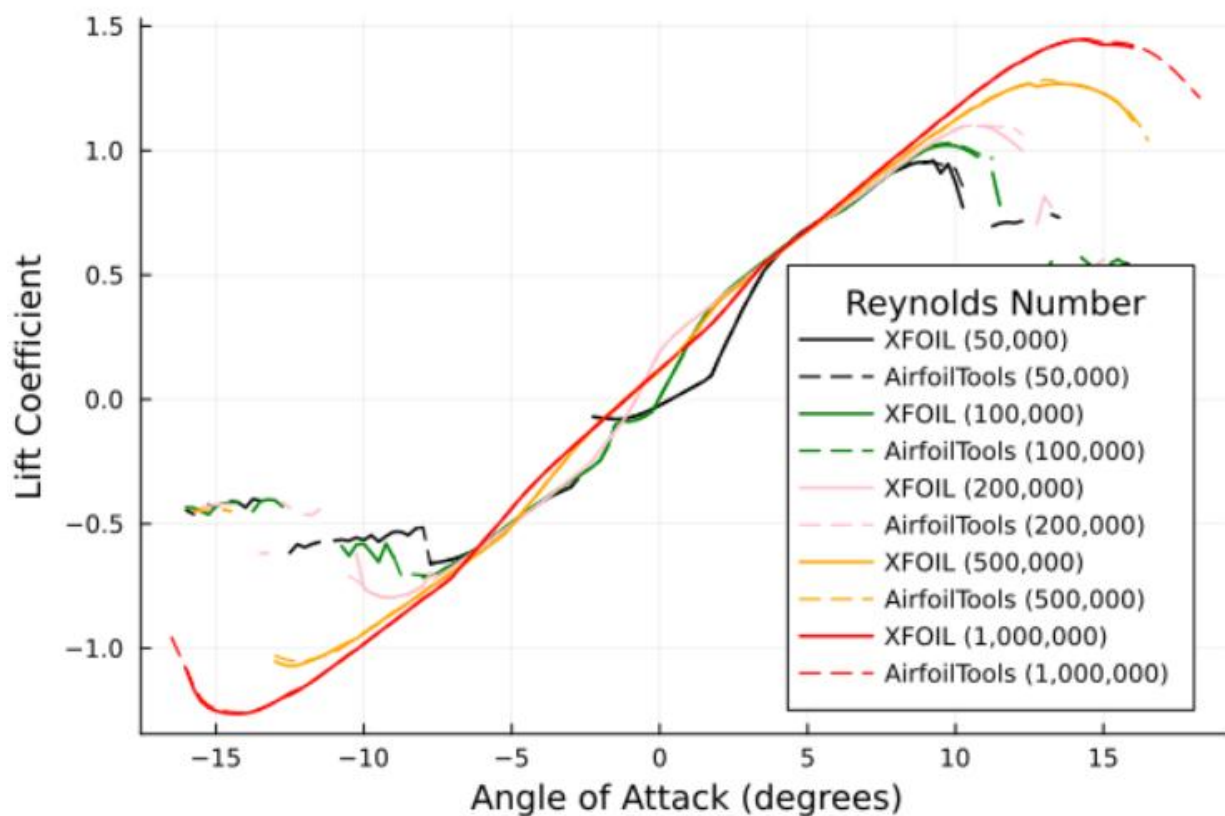
Introduction

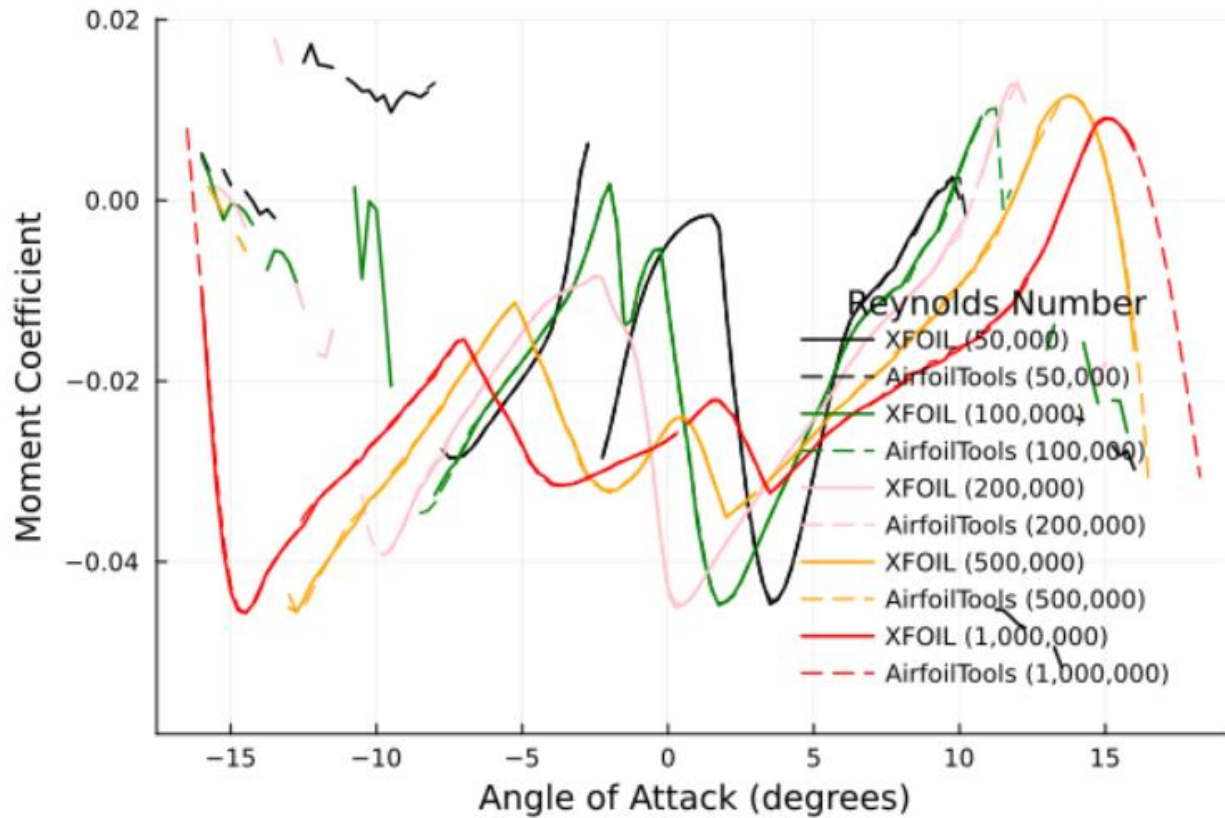
In this paper I am going to use Xfoil to predict the behavior of different airfoils, and how different Reynolds numbers affect their coefficient of lift, drag and moment graphs. Then I am going to compare that data to experimental or other data available on online databases.

I am going to first quote experimental or other data available, followed by the data I obtained from Xfoil:

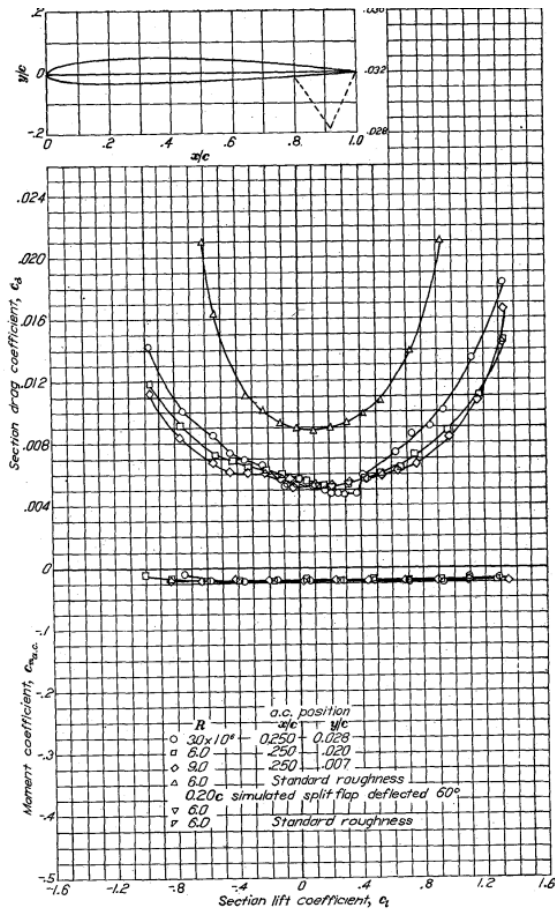
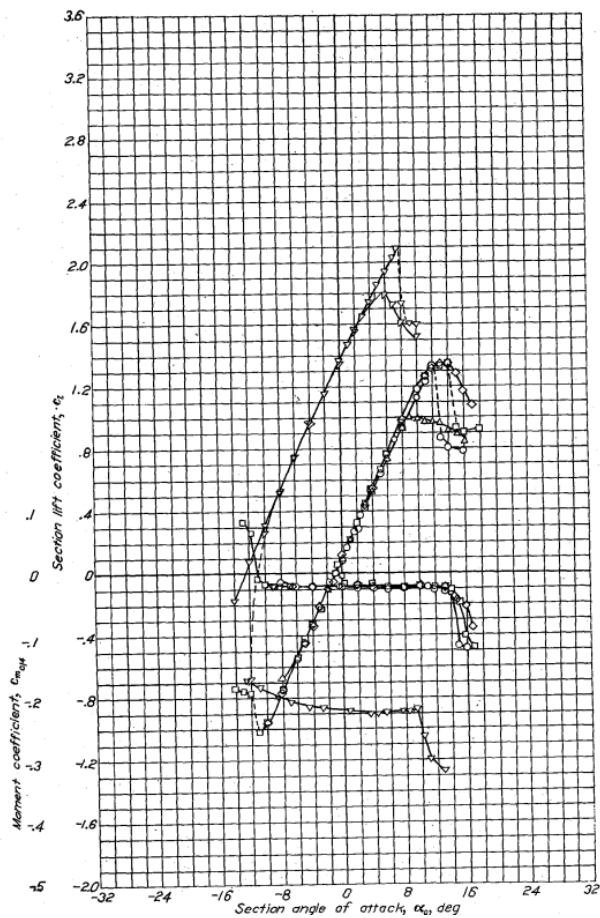
NACA 1410 reference plots





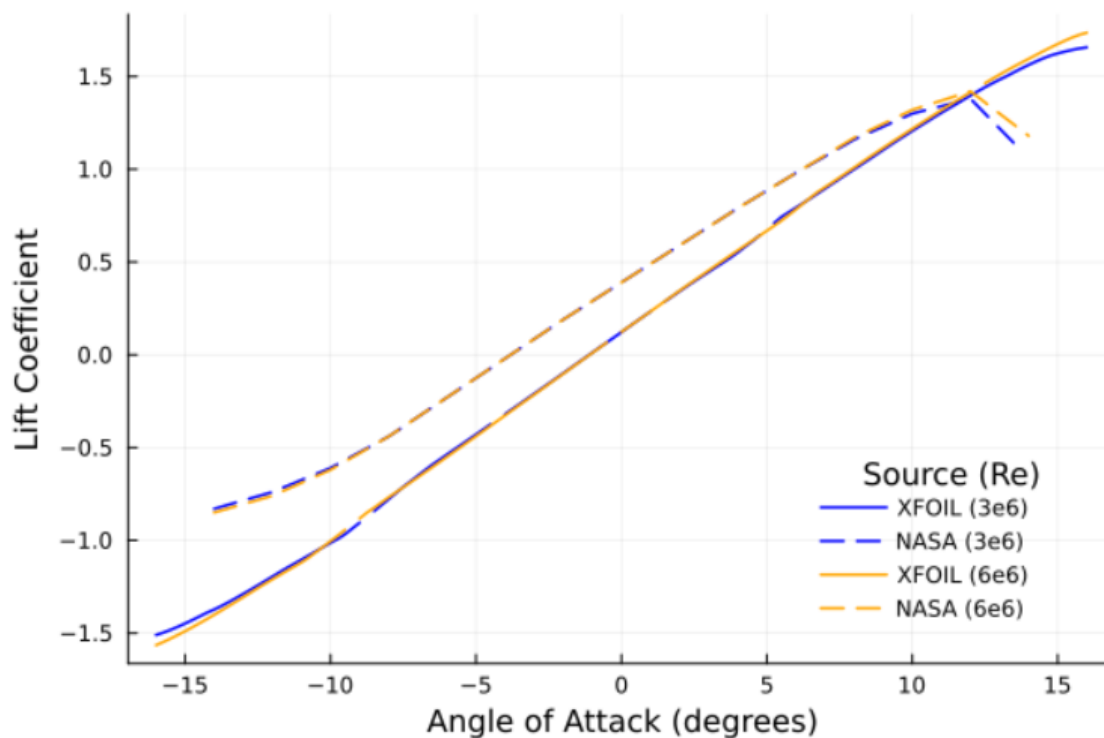
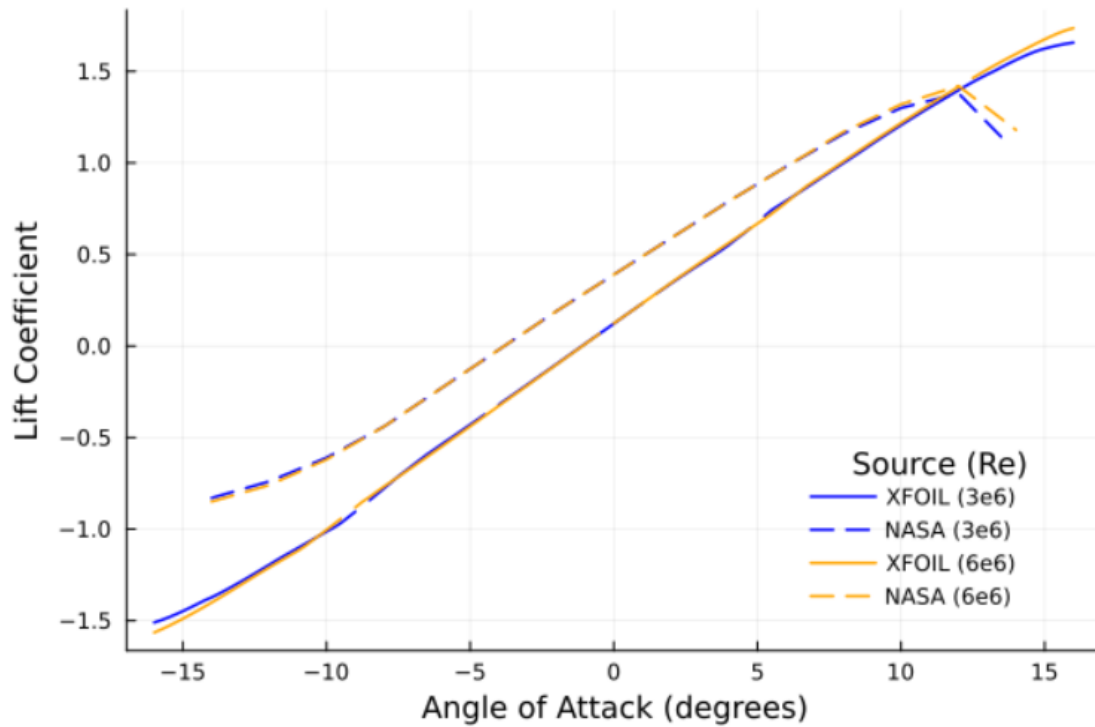


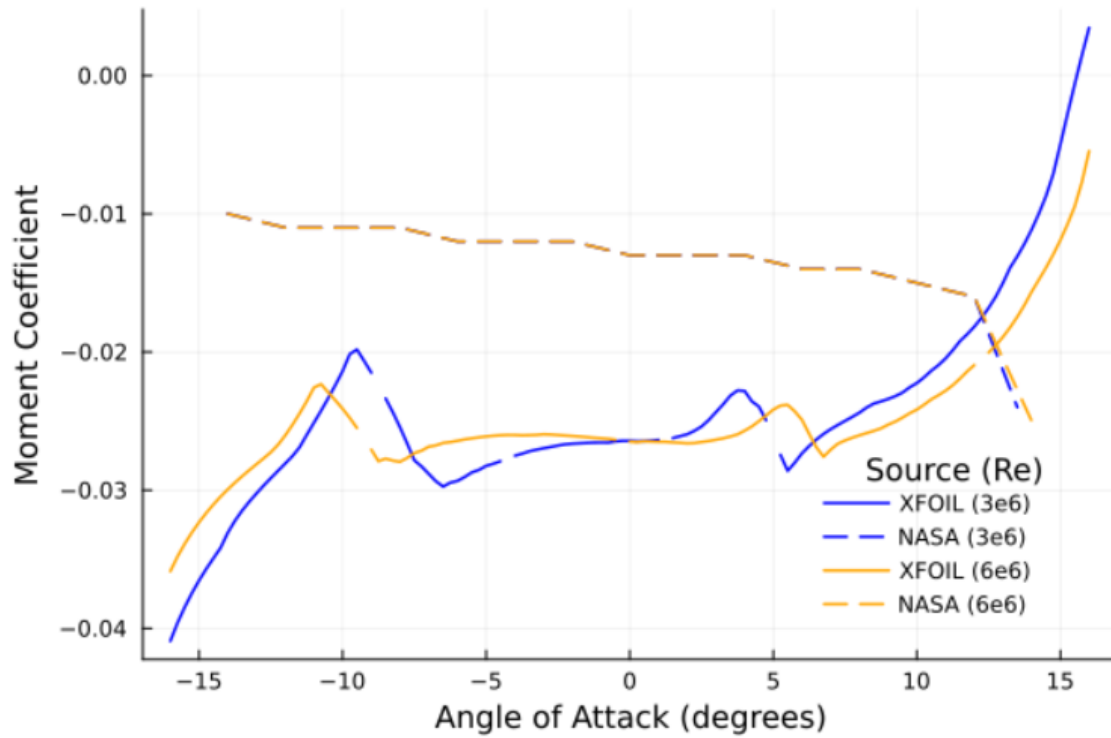
I created a filter for values that were physically suspicious and made the curves non-smooth, however, airfoiltools.com includes tables with their data, but they did not apply such filters. As expected, my values are exactly the same as airfoiltools.com, since they also used the Xfoil package, however, those out-of-range values are visible for airfoiltools.com lines distinguished by dashes.



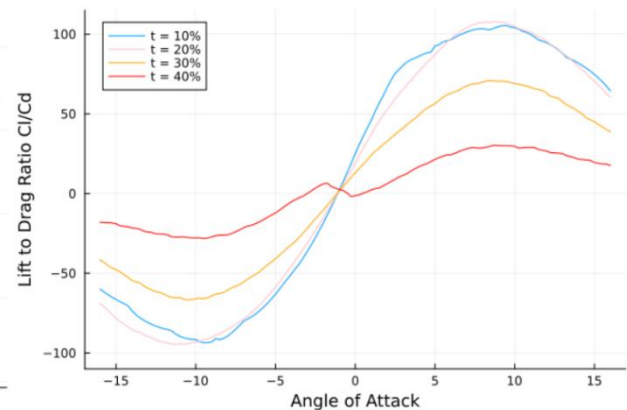
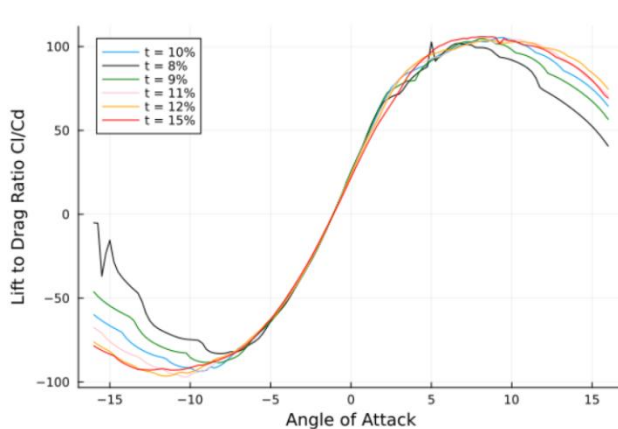
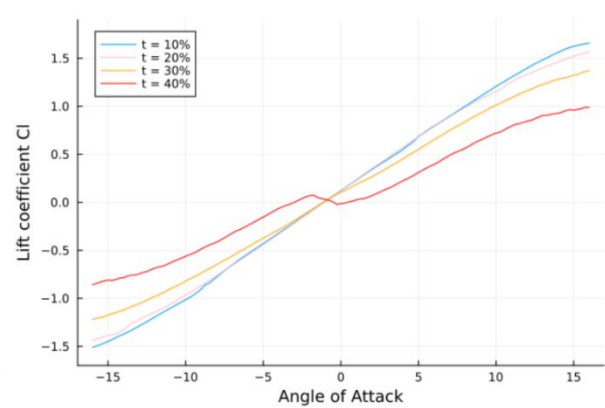
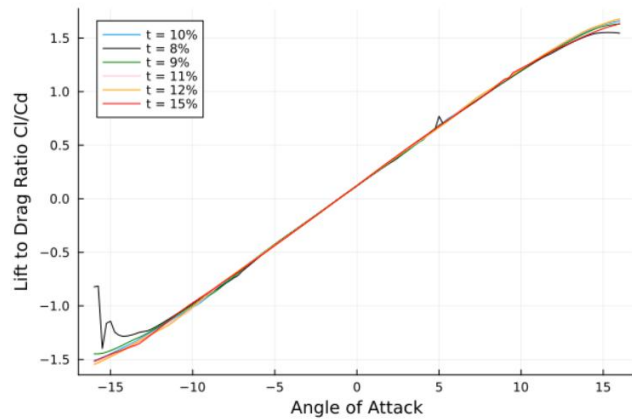
NACA 1408

Reynolds number: 3,000,000 and 6,000,000 (clean and smooth. NASA represents that line with squares)

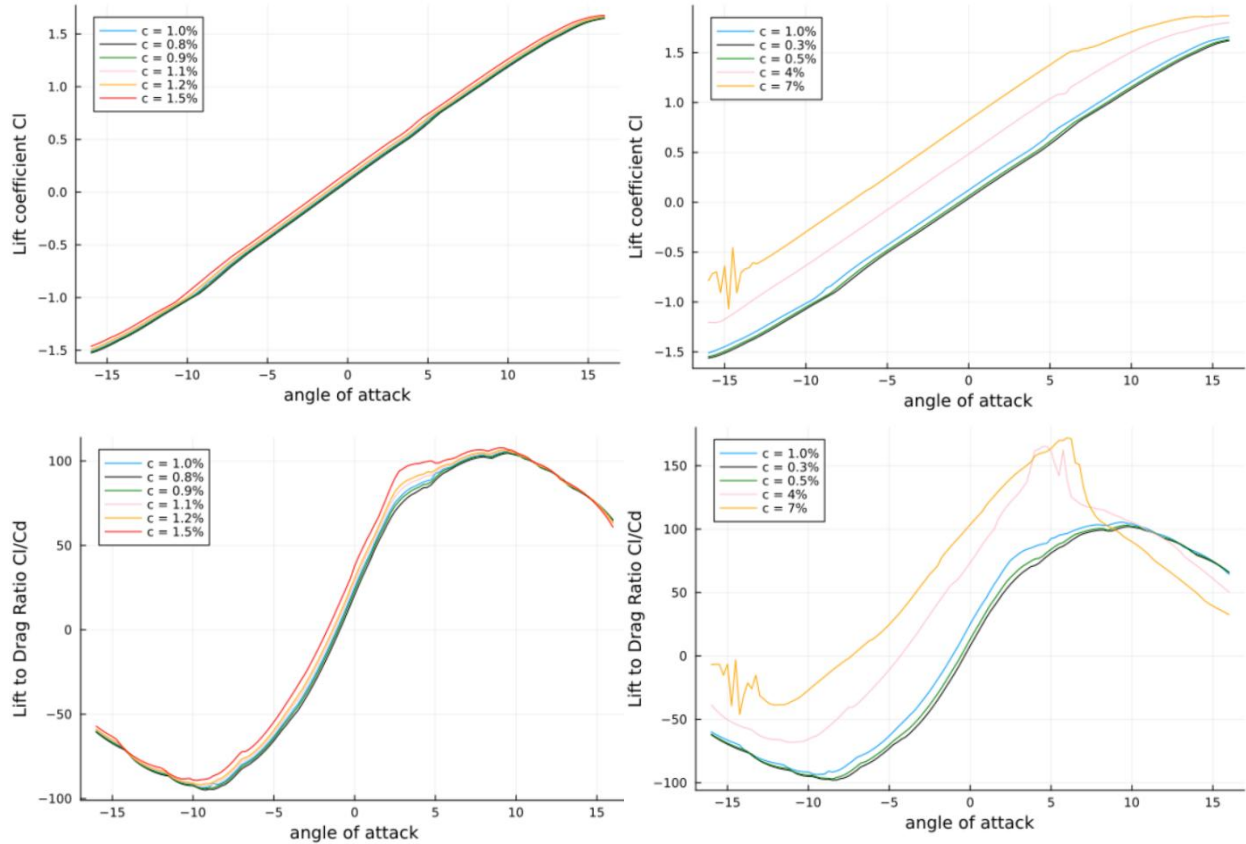




Varying thickness (keeping Reynolds the same):



Varying max Camber line (keeping Reynolds the same):



Methods:

Airfoil:

I used Xfoil to predict the behavior of airfoil NACA 1410, which has an effective max camber of 0.97% of its chord.

Reference data:

Xfoil coefficients calculated by airfoiltools.com and experimental data compiled by NASA on the NACA 1410

Coding methods:

I used the Xfoil tool created by Mark Darelà to analyze this airfoil using its *pane* and *solve_alpha* functions. Then plotted them using the Plots package. I used the same Reynolds numbers used in experimental and airfoiltools.com data. It is important to note that Xfoil's predictions were significantly close to the experimental.

Varying thicknesses:

For the varying thickness method, I started with variations of no more than 30%, however these did not show significant differences on the coefficient of lift graph. Conversely, the Coefficient to drag ratio graph did show differences before negative 10 and after positive 10 degrees. Because of these small variations I decided to use varying thicknesses with differences of over 400%

I learned how to use xfoil and **for** with variables with tuples inside to run the function I created to vary the thickness, and to plot such results

Varying camber line:

To alter the camber line, I had to:

1. Calculate the coordinates of the mean camber line
2. Find its maximum value
3. Use the max value to create a ratio with the new value I wanted to use
4. Multiply each coordinate of the original camber line by this ratio
5. Recalculate each coordinate of the new airfoil with altered camber line

Just as I did with varying thickness, I used **for** iterations with variables containing.

Results

Xfoil predictions of the base airfoil coordinates (initial test were airfoil shape remained unchanged) were very close to experimental data provided by NASA, with minimum variations, as the graphs show. Notice that the experimental data uses higher Reynolds' number than airfoiltools.com, hence I did different graphs comparing each.

It is important to note that NASA's coefficient of moment for Reynolds 3e6 and 6e6 are almost the same, and significantly different to Xfoil predictions.

As expected, my Xfoil predictions match airfoiltools.com because they also used Xfoil.

The most interesting part comes when varying thickness and max camber separately. The original thickness of the NACA 1410 is 10%. I tested different thicknesses keeping the camber line in place, not changing thickness by more than 1.5 times its original value. In these experiments, varying thickness did not have a significant impact on lift, but it did on drag. The results suggest that the thicker the airfoil the greater the drag only for lower values than -10 and higher than 10 degrees angle of attack.

Due to the small differences in thickness, I decided to vary it by 4 times its original value at most. Results indicate that lift increases significantly for negative angles of attack but decreases significantly for positive angles of attack.

I ran the same procedure with camber line. Kept the original thickness the same, varying the max value for camber line keeping it at the same distance from the leading edge. Results were like those of the original airfoil. I then decided to increase the variations by at most 7 times its original max value. The data suggests that, at a greater max value for camber line, lift is greater as well.

Final words

My takeaway is what I learned calculating all the values needed for this research activity. The use of **for** loops and variables with tuples, and the use of such a useful tool as Xfoil, as well as using the Plots package effectively.